

# Combinatorics

## Lecture 8

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We have already seen (in the case of Catalan, Fibonacci or Stirling numbers) that relations between numbers in a sequence may fully characterize the sequence (if some starting value is given).

A sequence  $(a_n)$  is defined recursively if

$a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k})$  for some function  $f$ , a natural  $k$ , and all  $n \geq 1$ . This identity  $a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k})$  is called a **recurrence relation** of order  $k$ . If  $a_0, a_1, \dots, a_{k-1}$  are known, they are called initial conditions.

The sequence  $(a_n)$  is then called a solution of the recursion  $f$  with initial values  $a_0, a_1, \dots, a_{k-1}$ . In the following we examine special cases of such recurrence relations and show how the sequence can be derived from them.

Any recurrence relation of the form

$$x_n = ax_{n-1} + bx_{n-2}$$

is called a **second order homogeneous linear recurrence relation**.

Let  $x_n = s_n$  and  $x_n = t_n$  be two solutions, then for constants  $c_1, c_2$

$$\begin{aligned} c_1 s_n + c_2 t_n &= c_1 (as_{n-1} + bs_{n-2}) + c_2 (at_{n-1} + bt_{n-2}) = \\ &= a(c_1 s_{n-1} + c_2 t_{n-1}) + b(c_1 s_{n-2} + c_2 t_{n-2}) \end{aligned}$$

This means that  $x_n = c_1 s_n + c_2 t_n$  is also a solution. Thus

### Theorem 1.

Any linear combination of solutions of a homogeneous recurrence linear relation is also a solution.

In solving the first order homogeneous recurrence linear relation  $x_n = ax_{n-1}$  it is clear that the general solution is  $x_n = a^n x_0$ .

This means that  $x_n = a^n$  is a solution. This suggests that, for the second order homogeneous recurrence linear relation, we may have the solutions of the form  $x_n = r^n$ .

Indeed, put  $x_n = r^n$  into initial relation. We have

$$r^{n-2}(r^2 - ar - b) = 0$$

Thus either  $r = 0$  or  $r^2 - ar - b = 0$ .

The previous equation is called the **characteristic equation**.

## Theorem 2.

If the characteristic equation has two distinct roots  $r_1$  and  $r_2$ , then the general solution for our relation is given by

$$x_n = c_1 r_1^n + c_2 r_2^n$$

If the characteristic equation has only one root  $r$ , then the general solution is given by

$$x_n = c_1 r^n + c_2 n r^n$$

Proof. When the characteristic equation has two distinct roots  $r_1$  and  $r_2$  it is clear that both  $x_n = r_1^n$  and  $x_n = r_2^n$  are solutions, so are their linear combinations.

Now assume that characteristic equation has only one root  $r$ . Then

$$b = \frac{-a^2}{4} \quad \text{and} \quad r = \frac{a}{2}$$

Nextly we verify that  $x_n = n r^n$  is a solution. In fact,

$$\begin{aligned}
 ax_{n-1} + bx_{n-2} &= a(n-1)\left(\frac{a}{2}\right)^{n-1} + \left(\frac{-a^2}{4}\right)(n-2)\left(\frac{a}{2}\right)^{n-2} = \\
 &= n\left(\frac{a}{2}\right)^n = x_n \blacksquare
 \end{aligned}$$

Example 1 (Again Fibonacci!) Find a general formula for the Fibonacci sequence  $f_n = f_{n-1} + f_{n-2}$ ;  $f_0 = 0, f_1 = 1$ .

The characteristic equation  $r^2 - r - 1 = 0$  has two distinct roots

$$r_1 = \frac{1 + \sqrt{5}}{2} \quad \text{and} \quad r_2 = \frac{1 - \sqrt{5}}{2}$$

The general solution is given by

$$f_n = c_1 \left( \frac{1 + \sqrt{5}}{2} \right)^n + c_2 \left( \frac{1 - \sqrt{5}}{2} \right)^n$$

$$\begin{cases} c_1 + c_2 = 0 \\ c_1 \left( \frac{1+\sqrt{5}}{2} \right) + c_2 \left( \frac{1-\sqrt{5}}{2} \right) = 1 \end{cases}$$

We have  $c_1 = -c_2 = \frac{1}{\sqrt{5}}$ . Thus

$$f_n = \frac{1}{\sqrt{5}} \left( \frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left( \frac{1-\sqrt{5}}{2} \right)^n$$

Example 2. Find the solution for the recurrence relation

$x_n = 6x_{n-1} - 9x_{n-2}$ , where  $x_0 = 2, x_1 = 3$ .

The characteristic equation  $r^2 - 6r + 9 = 0$  has only one root  $r = 3$ . Then the general solution is

$$x_n = c_1 3^n + c_2 n 3^n$$

The initial conditions imply that  $c_1 = 2$  and  $c_2 = -1$ . Thus the solution is

$$x_n = 2 \cdot 3^n - n \cdot 3^n = (2 - n)3^n$$

Let's consider a more general situation.

For a **higher order homogeneous recurrence relation**

$$x_{n+k} = a_1 x_{n+k-1} + a_2 x_{n+k-2} + \dots + a_{n-k} x_n$$

we also have the characteristic equation

$$t^k - a_1 t^{k-1} - a_2 t^{k-2} - \dots - a_{n-k+1} t - a_{n-k} = 0$$

As in the previous case, we get

### Theorem 3.

For the higher recurrence relation, if its characteristic equation has distinct roots  $r_1, r_2, \dots, r_k$ , then the general solution is

$$x_n = c_1 r_1^n + c_2 r_2^n + \dots + c_k r_k^n$$

where  $c_1, c_2, \dots, c_k$  are arbitrary constants. If the characteristic equation has repeated roots  $r_1, r_2, \dots, r_s$  with multiplicities  $m_1, m_2, \dots, m_s$  respectively, then the general solution is a linear combination of the solutions

$$r_1^n, nr_1^n, \dots, n^{m_1-1} r_1^n$$

$$r_2^n, nr_2^n, \dots, n^{m_2-1} r_2^n$$

.....

$$r_s^n, nr_s^n, \dots, n^{m_s-1} r_s^n$$

Example 3. Find an explicit formula for the sequence given by the recurrence relation

$$\begin{cases} x_n = 15x_{n-2} - 10x_{n-3} - 60x_{n-4} + 72x_{n-5} \\ x_0 = 1, x_1 = 6, x_2 = 9, x_3 = -110, x_4 = -45 \end{cases}$$

Proof. The characteristic equation is

$r^5 - 15r^3 + 10r^2 + 60r - 72 = 0$  There are roots  $r_1 = 2$  with multiplicity 3 and  $r_2 = -3$  with multiplicity 2. The general solution is given by

$$x_n = c_1 2^n + c_2 n 2^n + c_3 n^2 2^n + c_4 (-3)^n + c_5 n (-3)^n$$

The initial condition means that

$$\begin{cases} c_1 + c_4 = 1 \\ 2c_1 + 2c_2 + 2c_3 - 3c_4 - 3c_5 = 1 \\ 4c_1 + 8c_2 + 16c_3 + 9c_4 + 18c_5 = 1 \\ 8c_1 + 24c_2 + 72c_3 - 27c_4 - 81c_5 = 1 \\ 16c_1 + 64c_2 + 256c_3 + 81c_4 + 324c_5 = 1 \end{cases}$$

Solving the linear system we have

$$c_1 = 2, c_2 = 3, c_3 = -2, c_4 = -1, c_5 = 1$$

Thus the solution is

$$x_n = 2^{n+1} + 3n2^n - 2n^22^n - (-3)^n + n(-3)^n$$

A recurrence relation of the form  $x_n = ax_{n-1} + bx_{n-2} + f(n)$  is called a **non-homogeneous recurrence relation**.

Let  $x_n^{(s)}$  be a solution, called a **special solution**. Then the general solution is  $x_n = x_n^{(s)} + x_n^{(h)}$ , where  $x_n^{(h)}$  is the general solution for the corresponding homogeneous recurrence relation

$$x_n = ax_{n-1} + bx_{n-2}.$$

#### Theorem 4.

Let  $f(n) = cr^n$ . Let  $r_1$  and  $r_2$  be the roots of the characteristic equation  $t^2 - at - b = 0$ .

- (a) If  $r \neq r_1, r \neq r_2$ , then  $x_n^{(s)} = Ar^n$
- (b) If  $r = r_1, r_1 \neq r_2$ , then  $x_n^{(s)} = Anr^n$
- (c) If  $r = r_1 = r_2$ , then  $x_n^{(s)} = An^2r^n$

where  $A$  is a constant to be determined in all cases.

Proof. We assume  $r \neq 0$ . Otherwise the recurrence relation is homogeneous.

(a) Put  $x_n = Ar^n$  into relation. We have

$$Ar^n = aAr^{n-1} + bAr^{n-2} + cr^n$$

Thus

$$A(r^2 - ar - b) = cr^2.$$

Since  $r$  is not a root of the characteristic equation, then

$$r^2 - ar - b \neq 0$$

Hence

$$A = \frac{cr^2}{r^2 - ar - b}$$

(b) Since  $r = r_1 \neq r_2$ , it is clear that  $x_n = nr^n$  is not a solution for its corresponding homogeneous equation, i.e.,

$$nr^2 - a(n-1)r - b(n-2) = n(r^2 - ar - b) + ar + 2b = ar + 2b \neq 0$$

Put  $x_n = Anr^n$  into non-homogeneous recurrence relation. We have

$$Anr^n = aA(n-1)r^{n-1} + bA(n-2)r^{n-2} + cr^n$$

Thus  $A(nr^2 - a(n-1)r - b(n-2)) = cr^2$ . Therefore  $A = \frac{cr^2}{ar+2b}$ .

(c) Since  $r = r_1 = r_2$ , then  $a^2 + 4b = 0$ ,  $r = \frac{a}{2}$ , and  $x_n = n^2 r^n$  is not a solution of the corresponding homogeneous equation, i.e.,

$$\begin{aligned} n^2 r^2 - a(n-1)^2 r - b(n-2)^2 &= \\ = n^2(r^2 - ar - b) + 2n(ar + 2b) - ar - 4b &= -ar - 4b \neq 0 \end{aligned}$$

Put  $x_n = An^2r^n$  into initial relation. We have

$$Ar^{n-2}(n^2r^2 - a(n-1)^2r - b(n-2)^2) = cr^n$$

Thus  $A = \frac{-cr^2}{ar+4b}$ . ■

Example 4. Consider the non-homogeneous equation

$$\begin{cases} x_n = 3x_{n-1} + 10x_{n-2} + 7 \cdot 5^n \\ x_0 = 4, x_1 = 3 \end{cases}$$

The characteristic equation is  $t^2 - 3t - 10 = 0$ . We have roots  $r_1 = 5$ ,  $r_2 = -2$ . Since  $r = 5$ , then  $r = r_1$  and  $r \neq r_2$ . A special solution can be of the type  $x_n = An5^n$ . Put the solution into the non-homogeneous relation. We have

$$An5^n = 3A(n-1)5^{n-1} + 10A(n-2)5^{n-2} + 7 \cdot 5^n$$

Dividing both sides by  $5^{n-2}$ ,

$$An5^2 = 3A(n-1)5 + 10A(n-2) + 35$$

Thus  $A = 5$ , so  $x_n = n5^{n+1}$ . The general solution is

$$x_n = n5^{n+1} + c_15^n + c_2(-2)^n$$

The initial condition implies  $c_1 = -2$  and  $c_2 = 6$ . Therefore

$$x_n = n5^{n+1} - 2 \cdot 5^n + 6(-2)^n$$

Example 5. Compute the following determinant

$$\Delta_n = \begin{vmatrix} 3 & 2 & 0 & 0 & \dots & 0 \\ 1 & 3 & 2 & 0 & \dots & 0 \\ 0 & 1 & 3 & 2 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 3 & 2 \\ 0 & \dots & 0 & 0 & 1 & 3 \end{vmatrix}.$$

Рис.: TriDet

Expanding the determinant on the first line, we obtain the recurrence relation:

$$\Delta_n = 3\Delta_{n-1} - 2\Delta_{n-2}$$

The characteristic equation is  $t^2 - 3t + 2 = 0$ , which has roots  $t_1 = 1$ ,  $t_2 = 2$ .

According to Theorem 2, the general solution is

$$\Delta_n = C_1 2^n + C_2$$

Let's calculate constants  $C_1$  and  $C_2$  for  $n = 1, 2$ :

$$\begin{cases} 2C_1 + C_2 = 3 \\ 4C_1 + C_2 = 7 \end{cases}$$

Thus, we obtain  $C_1 = 2$ ,  $C_2 = -1 \implies \Delta_n = 2^{n+1} - 1$

Example 6. Let  $S = \{a_1, a_2, \dots, a_n\} \subset \mathbb{Z}$ , where  $n = 2^k$ . How many number of comparisons are needed in worst case to find the minimum in  $S$ ?

If all possible pairs of elements in  $S$  are compared, then the minimum will be found, and the number of comparisons in worst case is  $C_n^2$ . Of course this is not best possible.

There is another method to find a better solution. Let  $x_n$  be the least number of comparisons needed in worst case to find the minimum in  $S$ . Obviously,  $x_1 = 0$  and  $x_2 = 1$ . For  $n = 2^k$  we may divide  $S$  into two subsets

$$S_1 = \{a_1, a_2, \dots, a_{\frac{n}{2}}\}, \quad |S_1| = \frac{n}{2}$$

$$S_2 = \{a_{\frac{n}{2}+1}, a_{\frac{n}{2}+2}, \dots, a_n\}, \quad |S_2| = \frac{n}{2}$$

It takes  $x_{\frac{n}{2}}$  comparisons to find the minimum  $m_1$  for  $S_1$  and the minimum  $m_2$  for  $S_2$ . Then compare  $m_1$  with  $m_2$  to determine the minimum in  $S$ . In this way the total number of comparisons for  $S$  in worst case is  $2x_{\frac{n}{2}} + 1$ . We thus obtain a recurrence relation

$$\begin{cases} x_n = 2x_{\frac{n}{2}} + 1 \\ x_2 = 1 \end{cases}$$

Applying the recurrence relation iteratively, we have

$$\begin{aligned} x_n &= 2(2x_{\frac{n}{2^2}} + 1) + 1 = 2^2x_{\frac{n}{2^2}} + 2 + 1 = 2^2(2x_{\frac{n}{2^3}} + 1) + 2 + 1 = \\ &= 2^3x_{\frac{n}{2^3}} + 2^2 + 2 + 1 = \dots = 2^{k-1}x_{\frac{n}{2^{k-1}}} + \dots + 2^2 + 2 + 1 = \\ &= 2^{k-1} + \dots + 2^2 + 2 + 1 = 2^k - 1 = n - 1. \end{aligned}$$